

Better Policies from Short Comparisons: Minimal Post-Training with Frozen World Models

Anonymous submission

Abstract

World-model agents are typically trained under fixed interaction and compute budgets, so training may stop before the policy fully exploits the final world model. A natural remedy is to freeze the model and continue imagination-based actor-critic updates, but this is often unreliable: as the policy moves away from the source policy, model errors can compound and be exploited. We find that a weaker signal remains reliable after training: short branches from the same latent state often preserve the relative ranking of alternative first actions, even when their absolute return estimates are poorly calibrated. Based on this observation, we introduce Comparative Policy Repair (CPR), a lightweight post-training method that uses the frozen world model as a local action comparator rather than a long-horizon simulator. CPR compares short branches that differ only in the first action, follows the frozen source policy thereafter, combines each branch's short-horizon return with an action-conditioned Q estimate, and distills the resulting relative advantages into the actor under a KL constraint to the source policy. Across all settings covering Atari 100K, DMC-Proprio, and Meta-World MT50 with different world-model agents, CPR consistently improves performance. It yields **3.8%–5.2%** aggregate gains on Atari and DMC and a **5.0**-percentage-point increase in MT50 success rate, with no additional environment interaction and only **0.1%–0.7%** of the original training compute.

Introduction

Model-based reinforcement learning (MBRL) improves sample efficiency by learning predictive models and optimizing policies through imagined experience (Sutton 1990; Ha and Schmidhuber 2018). Recent latent world models train strong agents from limited interaction by learning compact dynamics and improving policies in imagination (Hafner et al. 2019, 2023; Peng et al. 2026; Burchi and Timofte 2025; Micheli, Alonso, and Fleuret 2023; Alonso et al. 2024; Zhang et al. 2026). Yet fixed interaction and compute budgets do not guarantee that the final policy fully exploits the final model.

Imagination-based agents typically alternate environment interaction, world-model learning, and actor-critic optimization (Hafner et al. 2023; Micheli, Alonso, and Fleuret 2023). Their policies are thus optimized against a sequence of evolving models, rather than fully re-optimized against the final model after training ends. Once the budget is exhausted, continuing the standard online loop requires additional inter-

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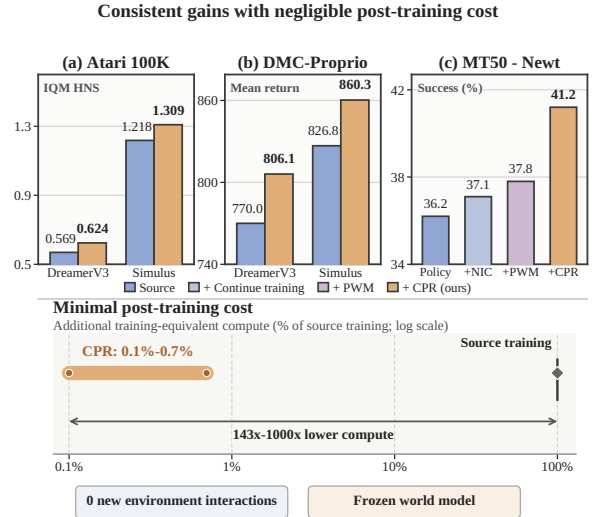


Figure 1: **Five-seed aggregate performance and post-training cost.** Top: Atari IQM human-normalized score, DMC mean return, and MT50 success. Bottom: CPR uses 0.1%–0.7% of source-training compute, no new environment interaction, and a frozen world model.

action and model updates. Figure 1 shows measurable post-training headroom across Atari 100K, DMC-Proprio, and MT50: the source policies can still be improved after training. For Atari and DMC, we quantify this headroom with Mean Relative Gain, whose definition is given in Appendix *Experimental Setup and Reproducibility*.

A natural alternative is to freeze the final world model and continue the original imagined actor-critic updates, which we call Frozen-WM Continue. However, as the policy drifts from the model's training distribution, it may visit poorly modeled state-action regions and exploit inaccurate predictions, while model errors accumulate over long rollouts (Talvitie 2017; Janner et al. 2019; Yu et al. 2020b, 2021). Although long rollouts drift, short same-state branches can still preserve useful rankings among alternative first actions. CPR asks a simple question at each replayed state: which first action produces the best short branch when all later actions

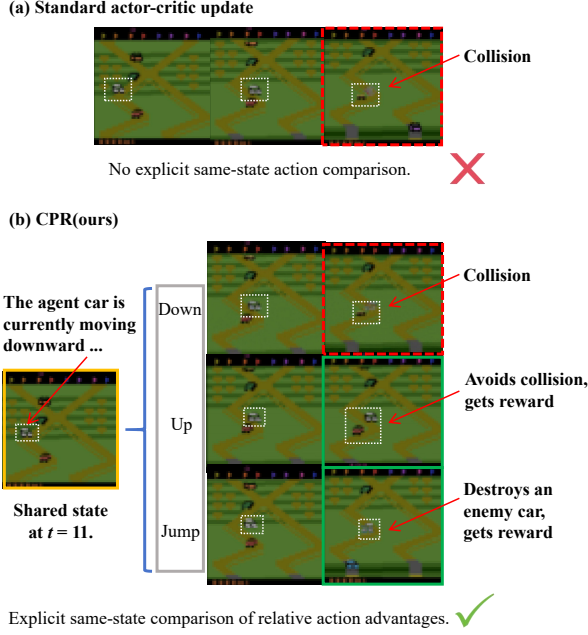


Figure 2: **Same-state action comparison.** CPR varies only the first action at a shared state; all later actions are sampled from π_0 . The short branches provide relative action rankings.

are sampled from the same frozen source policy?

We introduce **Comparative Policy Repair (CPR)**, a post-training method that exploits this local ranking signal through short branching, Q -based branch scoring, and a source-constrained actor update. From each latent state, CPR samples candidate first actions, expands short branches in the frozen model, estimates their relative advantages, and updates the actor through source-weighted regression with a KL constraint to the source policy. CPR leaves the world model unchanged and requires no additional training interaction. Across Atari 100K and DMC-Proprio, CPR yields positive aggregate improvements in all four settings; on Meta-World MT50, it improves success by 5.0 percentage points. These gains require only 0.1%–0.7% of source-training compute.

Our contributions are summarized as follows:

- Short same-state branches retain action-ordering information after their absolute returns begin to drift.
- CPR combines branch scoring with source-weighted, KL-regularized actor updates while freezing the world model.
- Across five benchmark–backbone combinations, CPR improves performance using 0.1%–0.7% of source-training compute and no new environment interaction.

Background

We consider a partially observable Markov decision process (POMDP) (Kaelbling, Littman, and Cassandra 1998) with observations o_t , actions a_t , rewards r_t , and continuation flags c_t . The policy maximizes the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t]$.

Learning in World Models

Model-based reinforcement learning improves sample efficiency by learning an action-conditioned predictive model of the environment (Ha and Schmidhuber 2018; Hafner et al. 2019). Learned world models support decision-time planning (Schrittwieser et al. 2020; Ye et al. 2021) or policy optimization from imagined experience (Hafner et al. 2020, 2021, 2023; Micheli, Alonso, and Fleuret 2023).

We use a generic latent-state formulation that covers different world model architecture. We denote the predictive components by $\mathcal{M}_\phi = \{E_\phi, T_\phi, R_\phi, C_\phi\}$, with an optional observation decoder D_ϕ . The complete history required for prediction is encoded into a model state

$$z_t = E_\phi(o_{\leq t}, a_{< t}), \quad (1)$$

where z_t may include recurrent hidden states or cached sequence context. The predictive dynamics are written as

$$\hat{z}_{t+1} \sim T_\phi(\cdot | z_t, a_t), \quad (2)$$

$$\hat{r}_t = R_\phi(z_t, a_t), \quad (3)$$

$$\hat{c}_t = C_\phi(z_t, a_t). \quad (4)$$

Here $\hat{c}_t$ is the predicted probability that the episode continues; for a model that predicts termination, $\hat{c}_t = 1 - \hat{d}_t$. Some visual world models additionally reconstruct observations as $\hat{o}_t = D_\phi(z_t)$, whereas latent or implicit models need not decode observations (Hafner et al. 2023; Micheli, Alonso, and Fleuret 2023; Hansen, Su, and Wang 2024). Repeated application of Equations (2)–(4) generates imagined trajectories without additional environment interaction.

Rollout error grows with horizon, and policy can exploit inaccurate predictions as it moves away from the data distribution (Janner et al. 2019; Yu et al. 2020b, 2021). CPR therefore uses short same-state comparisons rather than treating each imagined trajectory as an absolute target (Figure 2).

Policy Optimization in Imagination

Imagination-based agents commonly optimize an actor π_θ with a critic or bootstrap estimator using imagined trajectories (Hafner et al. 2020, 2021, 2023). Standard policy-gradient updates estimate an advantage for the action sampled along each imagined trajectory, using Monte Carlo returns, temporal-difference residuals, generalized advantage estimation, or λ -returns (Williams 1992; Schulman et al. 2016; Sutton and Barto 2018). CPR instead compares several first actions from the same anchor state.

Method

We freeze the world model $\mathcal{M}_\phi$ and source policy π_0 , initialize $\pi_\theta \leftarrow \pi_0$, and train π_θ on the existing replay dataset $\mathcal{D}$ without new environment interaction. CPR has three stages: shared-anchor branching, Q -based branch scoring, and a KL-regularized weighted policy update. Throughout this section, superscripts index imagined branches: 0 denotes the source branch, while $k \in \{1, \dots, K\}$ indexes a candidate branch. Appendix *Additional Method Details* provides complete pseudocode, component freeze boundaries, branch-generation details, and evaluator interfaces for the three backbone families.

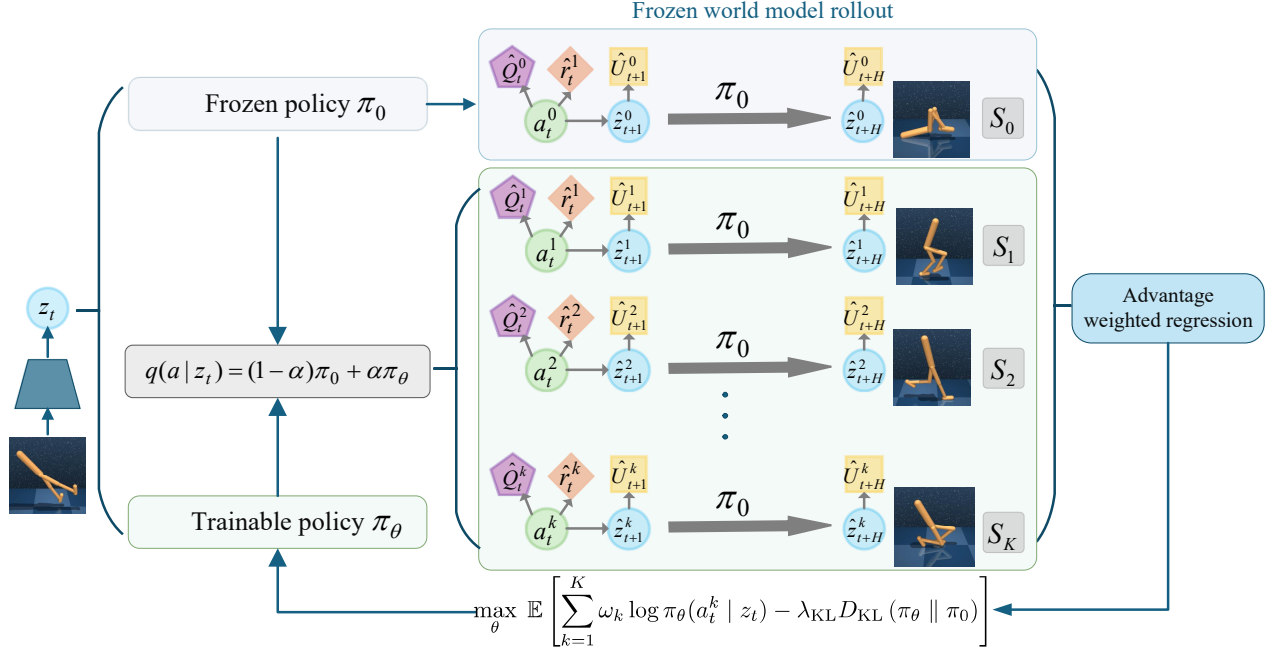


Figure 3: **Overview of CPR.** From a shared state, CPR samples candidate first actions, rolls them out under π_0 in the frozen model, scores them relative to the source branch, and fits π_θ using KL-regularized weighted regression.

Shared-Anchor Branching

Standard imagination-based agents typically generate trajectories by repeatedly sampling from a single policy. CPR instead queries multiple possible futures from the same model state. Given an anchor z_t sampled from $\mathcal{D}$, we construct the proposal distribution

$$q_\theta(a | z_t) = (1 - \alpha)\pi_0(a | z_t) + \alpha\pi_\theta(a | z_t), \quad (5)$$

where $\alpha \in [0, 1]$ controls the mixture of current- and source-policy proposals. We sample K candidate first actions $a_t^k \sim q_\theta(\cdot | z_t)$ for $k = 1, \dots, K$.

For each candidate, we initialize $\hat{z}_t^k = z_t$ and generate an H -step imagined branch. After the candidate first action, all candidate branches use the frozen source policy as their common continuation rule:

$$\begin{aligned} a_{t+i}^k &\sim \pi_0(\cdot | \hat{z}_{t+i}^k), & i &= 1, \dots, H-1, \\ \hat{z}_{t+i+1}^k &\sim T_\phi(\cdot | \hat{z}_{t+i}^k, a_{t+i}^k), & i &= 0, \dots, H-1, \\ \hat{r}_{t+i}^k &= R_\phi(\hat{z}_{t+i}^k, a_{t+i}^k), & i &= 0, \dots, H-1, \\ \hat{c}_{t+i}^k &= C_\phi(\hat{z}_{t+i}^k, a_{t+i}^k), & i &= 0, \dots, H-1. \end{aligned} \quad (6)$$

Within the candidate set, the first action is the only explicit intervention; all subsequent actions follow the same source-policy continuation rule. Equation (5) applies to both categorical and continuous policy distributions.

We also generate a source branch by sampling its first and later actions from π_0 . Candidate and source branches therefore differ only in the candidate first-action distribution; all continuations use π_0 in the same frozen world model.

Comparative Branch Evaluation

Finite-horizon returns are fragile to reward, continuation, and bootstrap errors, and may encourage exploitation of model inaccuracies (Janner et al. 2019; Yu et al. 2020b, 2021). CPR instead anchors each branch score with $Q_\psi(z_t, a_t^k)$ and adds short TD corrections along the continuation.

Action-conditioned evaluator. CPR accesses two scalar interfaces from the pretrained agent: an action-conditioned evaluator $Q_\psi(z, a)$ and a source-policy bootstrap $U_\xi(z)$. When the backbone provides a frozen N -head action-value ensemble $\{Q_{\psi_n}\}_{n=1}^N$, we define

$$\begin{aligned} Q_\psi(z, a) &= \frac{1}{N} \sum_{n=1}^N Q_{\psi_n}(z, a), \\ U_\xi(z) &= \mathbb{E}_{a \sim \pi_0(\cdot | z)} [Q_\psi(z, a)]. \end{aligned} \quad (7)$$

For a deterministic source policy, this reduces to $U_\xi(z) = Q_\psi(z, \pi_0(z))$. The ensemble and source policy remain frozen throughout post-training.

When a native action-conditioned evaluator is unavailable, we attach a lightweight residual head

$$Q_\psi(z_t, a_t) = \text{sg}(U_\xi(z_t)) + \Delta Q_\psi(z_t, a_t), \quad (8)$$

where $\text{sg}(\cdot)$ denotes stop-gradient and ΔQ_ψ is the residual correction. For both interfaces, write $Q_t^k := Q_\psi(z_t, a_t^k)$, including $k = 0$. Trainable residual evaluators use replay TD targets bootstrapped by U_ξ ; Appendix *Action-Conditioned Evaluator* gives the initialization and loss details.

Short-horizon branch correction. For each branch, define the cumulative predicted continuation probability

$$\hat{m}_t^k = 1, \quad \hat{m}_{t+i}^k = \prod_{j=0}^{i-1} \hat{c}_{t+j}^k, \quad i = 1, \dots, H. \quad (9)$$

For rollout steps after the candidate first action, the imagined temporal-difference residual is

$$\delta_{t+i}^k = \hat{r}_{t+i}^k + \gamma \hat{c}_{t+i}^k U_\xi(\hat{z}_{t+i+1}^k) - U_\xi(\hat{z}_{t+i}^k), \quad i = 1, \dots, H-1. \quad (10)$$

The resulting branch score is

$$S_k = Q_t^k + \sum_{i=1}^{H-1} \gamma^i \hat{m}_{t+i}^k \delta_{t+i}^k. \quad (11)$$

The sum begins at $i = 1$ because Q_t^k already includes the first transition (Sutton 1988). Since all later actions follow π_0 , S_k approximates the local source-policy value $Q^{\pi_0}(z_t, a_t^k)$, not an on-policy value under π_θ . The proposal mixture and KL regularizer keep updates near the regime where this approximation is meaningful. Applying the same construction with branch index 0 yields the source-branch score S_0 .

Relative branch advantage. We compare each candidate with both the local candidate set and the source behavior. The source-centered baseline is

$$b_t = \max\{S_0, \text{median}_{1 \leq j \leq K} S_j\}, \quad (12)$$

and the relative branch advantage is $A_k = S_k - b_t$. The median prevents an isolated high-scoring branch from defining the local baseline, while S_0 prevents updates toward actions that are better than other proposals but worse than the source-policy behavior.

Because score magnitudes vary across anchors, tasks, and model classes, we use a robust anchor-wise scale. Let

$$\begin{aligned} \mathcal{S}_t &= \{S_0, S_1, \dots, S_K\}, \\ \tilde{S}_t &= \text{median}(\mathcal{S}_t), \\ \text{MAD}(\mathcal{S}_t) &= \text{median}_{S \in \mathcal{S}_t} |S - \tilde{S}_t|. \end{aligned} \quad (13)$$

We then apply MAD robust scaling followed by signed-log compression:

$$\bar{A}_k = \text{sgn}(A_k) \log \left(1 + \frac{|A_k|}{\text{MAD}(\mathcal{S}_t) + \epsilon} \right), \quad (14)$$

where ϵ is a fixed numerical constant. This is robust to isolated score outliers, preserves the sign of the original relative advantage, and compresses large magnitudes without introducing an additional normalization hyperparameter.

Conservative Policy Projection

A high branch score does not by itself guarantee that the candidate first action is likely under the source behavior. To reduce policy drift into poorly modeled regions (Janner et al. 2019; Yu et al. 2020b, 2021), we downweight candidates with low source likelihood relative to the other proposals

at the same anchor. Let $\ell_k^0 = \log \pi_0(a_t^k | z_t)$ denote the source-policy log-probability or log-density of candidate k . We convert it into the relative source-likelihood weight

$$w_k^{\text{src}} = \exp \left(\ell_k^0 - \max_{1 \leq j \leq K} \ell_j^0 \right). \quad (15)$$

This definition applies to categorical probabilities and continuous-action densities. Since the normalization is anchor-wise, it should not be interpreted as certifying absolute behavioral support.

We distill the comparative signal using advantage-weighted regression (Peng et al. 2019; Nair et al. 2020). For each anchor, define

$$u_k = \mathbf{1}\{A_k > 0\} w_k^{\text{src}} \exp(\bar{A}_k / \beta), \quad (16)$$

$$w_k = \text{sg} \left(\frac{u_k}{\sum_{j=1}^K u_j + \epsilon} \right), \quad (17)$$

where β is the AWR temperature. The indicator is applied to the original advantage A_k , so only candidates scored above the source-centered baseline receive gradient. If no candidate has positive advantage, the anchor contributes no repair term.

The policy-repair objective is

$$\mathcal{L}_{\text{repair}} = -\mathbb{E}_{z_t \sim \mathcal{D}} \left[\sum_{k=1}^K w_k \log \pi_\theta(a_t^k | z_t) \right]. \quad (18)$$

We additionally constrain the updated policy to remain close to the frozen source policy:

$$\mathcal{L}_{\text{KL}} = \mathbb{E}_{z_t \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot | z_t) \| \pi_0(\cdot | z_t))]. \quad (19)$$

The policy is optimized with

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_{\text{repair}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}}. \quad (20)$$

All branch-derived states, scores, advantages, and weights are stop-gradient quantities; actor updates do not modify $\mathcal{M}_\phi$, π_0 , or the evaluator. Appendix *Common CPR Implementation* and Appendix *Backbone- and Benchmark-Specific Details* describe the concrete implementation choices for DreamerV3, Simulus, and Newt.

Experiments

We evaluate CPR across three benchmarks and three pre-trained world-model backbones, spanning discrete control, continuous control, and multi-task robotic manipulation.

Experimental Setup

Benchmarks. We evaluate on Atari 100K (26 pixel-based discrete-control games) (Kaiser et al. 2020; Bellemare et al. 2013), DMC-Proprio (18 continuous-control tasks with vector observations) (Tunyasuvunakool et al. 2020), and Meta-World MT50 (50 multi-task manipulation tasks) (Yu et al. 2020a; McLean et al. 2025). Atari agents use 100K interactions and report human-normalized scores (HNS) (Wang et al. 2016), including mean, median, interquartile mean (IQM), and optimality gap (Agarwal et al. 2021). DMC source agents use 500K interactions following TD-MPC2 (Hansen, Su, and Wang 2024) and report mean

Backbone	Method	Mean $\uparrow$	Median $\uparrow$	IQM $\uparrow$	Gap $\downarrow$	Gain (%) $\uparrow$	Imp. $\uparrow$
<i>Atari 100K (26 games; human-normalized scores)</i>							
DreamerV3	Source	1.089	0.526	0.569	0.463	–	–
DreamerV3	+NIC	1.083	0.506	0.555	0.474	-2.75	6/26
DreamerV3	+CPR (ours)	1.123	0.614	0.624	0.446	+4.70	19/26
Simulus	Source	2.101	0.823	1.218	0.404	–	–
Simulus	+NIC	2.075	0.800	1.174	0.405	-1.95	9/26
Simulus	+CPR (ours)	2.222	0.827	1.309	0.399	+5.15	21/26
<i>DMC-Proprio (18 tasks; episode returns)</i>							
DreamerV3	Source	770.0	880.7	–	–	–	–
DreamerV3	+NIC	754.7	894.6	–	–	-2.69	7/18
DreamerV3	+CPR (ours)	806.1	951.2	–	–	+4.78	15/18
Simulus	Source	826.8	918.4	–	–	–	–
Simulus	+NIC	788.3	851.8	–	–	-4.55	5/18
Simulus	+CPR (ours)	860.3	962.7	–	–	+3.84	15/18

Table 1: Five-seed aggregate results. Gain is the task-averaged relative change from Source; Imp. counts tasks with higher unrounded five-seed means. Best values are bold and CPR rows are shaded. On Atari and DMC, Continue is evaluated in the matched component-ablation curves in Figure 5, while this table reports the branch-budget-matched Source/NIC/CPR comparison. Appendix *Task-Level Result Schema and Artifacts* defines the task-level result artifact, aggregation protocol, and five-seed per-task means.

and median episode returns. MT50 follows the Newt setting (Hansen, Su, and Wang 2025) and reports mean success rate without test-time planning. All post-training uses only the replay buffers retained from source training. Appendix *Experimental Setup and Reproducibility* provides training, demonstration, and evaluation details, while Appendix *Task-Level Result Schema and Artifacts* defines the result artifact.

For Atari and DMC, Mean Relative Gain averages the per-task relative change from the corresponding source checkpoint after first averaging the five seeds. Improved Tasks counts tasks whose unrounded five-seed mean increases. The exact MRG definition, zero-denominator convention, rounding convention, and task-level aggregation rule are given in Appendix *Experimental Setup and Reproducibility*.

World-model backbones. CPR is evaluated on DreamerV3, a recurrent RSSM agent (Hafner et al. 2023); Simulus, a token-based world-model agent (Cohen et al. 2025); and Newt, an implicit multitask model with a pretrained Q ensemble (Hansen, Su, and Wang 2025). For Newt, CPR updates only the policy prior and uses no test-time planning.

Baselines and protocol. **Source** denotes the pretrained checkpoint. **Frozen-WM Continue (Continue)** resumes the backbone’s native policy-update objective while keeping the final world model frozen, testing whether additional policy optimization alone is sufficient. **Naive Imagined Continuation (NIC)** uses the same anchor states, candidate branches, and branch budget as CPR, but assigns uniform weights to candidate first actions and removes CPR’s comparative branch scoring and source-constrained update. On MT50, we additionally include a **PWM-style** baseline that directly optimizes the policy through predicted returns of the frozen model (Georgiev et al. 2025). **Newt-MPC** is included only as a test-time planning reference (Hansen, Su, and Wang 2025).

All actor-only methods start from the same source check-

point and replay data, use no new environment interaction, and are compute-matched within each direct comparison. Evaluation rollouts are never used for optimization, and all reported results average five seeds. On Atari and DMC, Continue is evaluated in the matched component-ablation curves in Figure 5, while Table 1 reports the branch-budget-matched Source/NIC/CPR comparison. On MT50, Frozen-WM Continue, PWM-style optimization, and CPR use the same frozen Newt model and replay data. CPR requires only 0.1%–0.7% of source-training compute. Update schedules, evaluation episodes, final CPR-specific hyperparameters, and implementation details are provided in Appendix *Experimental Setup and Reproducibility*.

Results and Analysis

Overall performance. Table 1 summarizes results on Atari 100K and DMC-Proprio, while Table 2 reports Meta-World MT50. Across the four Atari and DMC benchmark-backbone settings, CPR obtains mean relative gains of 3.84%–5.15%, improving performance for every source checkpoint. On MT50, CPR raises the success rate of the Newt policy prior from 36.2% to 41.2%, a gain of 5.0 percentage points, compared with 37.1% for Frozen-WM Continue and 37.8% for PWM-style optimization. These improvements require neither world-model updates nor additional training interaction. Appendix *Statistical Stability Analyses* reports stratified bootstrap confidence intervals, performance profiles, and paired probabilities of improvement; Appendix *Task-Level Result Schema and Artifacts* reports the five-seed per-task means.

Component ablations.

Validation of the core hypothesis. Figure 4 tests the premise behind CPR. From matched anchors, we execute

Table 2: Validation on Meta-World MT50 over 5 seeds. All actor-only post-training methods start from the same Newt-Policy checkpoint, use no additional training interaction, and are matched in post-training compute. “Continue” denotes Frozen-WM Continue. Newt-MPC[†] is included only as a test-time planning reference.

Method	Success (% , $\uparrow$)	Δ (pp)
Newt-Policy	36.2	—
+ Continue	37.1	+0.9
+ PWM-style	37.8	+1.6
+ CPR (ours)	41.2	+5.0
Newt-MPC [†]	55.6	+19.4

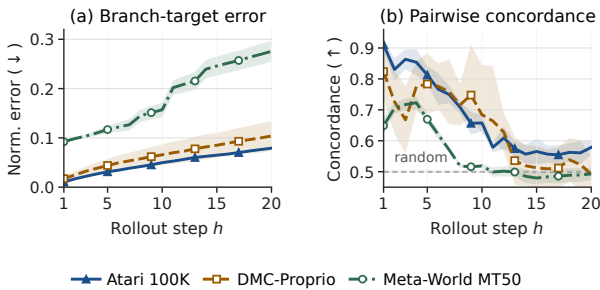


Figure 4: **Real-versus-model diagnostics from matched anchors.** Each branch starts from the same replay anchor, takes the same candidate first action in the real environment and frozen model, and then follows π_0 . Left: normalized λ -return error. Right: pairwise action-order concordance; the dashed line marks chance at 0.5. Shading denotes 95% confidence intervals. This diagnostic evaluates whether short branches preserve rankings, not whether raw branch returns are sufficient actor targets.

the same candidate first action in the real environment and the frozen world model, follow the source policy thereafter, and compare the resulting λ -return branch targets (Sutton 1988). Absolute λ -return error grows with horizon, but pairwise action concordance remains above chance over CPR’s short horizons and decays toward 0.5 at longer horizons, fastest on MT50. Thus, local rankings remain useful after absolute returns become poorly calibrated. This does not imply that raw branch returns are valid policy targets: replacing CPR’s Q -based score with the matched λ -return score degrades all four settings (Table 3). Appendix *Core-Hypothesis Validation Details* provides the matched-anchor protocol, real-versus-model branch comparison, and score-specific precision, coverage, and regret diagnostics for S_k , including positive-advantage precision and coverage audits that separate the λ -return diagnostic from CPR’s actual Q -based branch score.

The diagnostic is aligned with CPR’s intervention structure: candidate actions share the same anchor, differ only at the first decision, and use the same source-policy continuation afterward. This removes variation from later policy choices and tests whether the world model preserves the

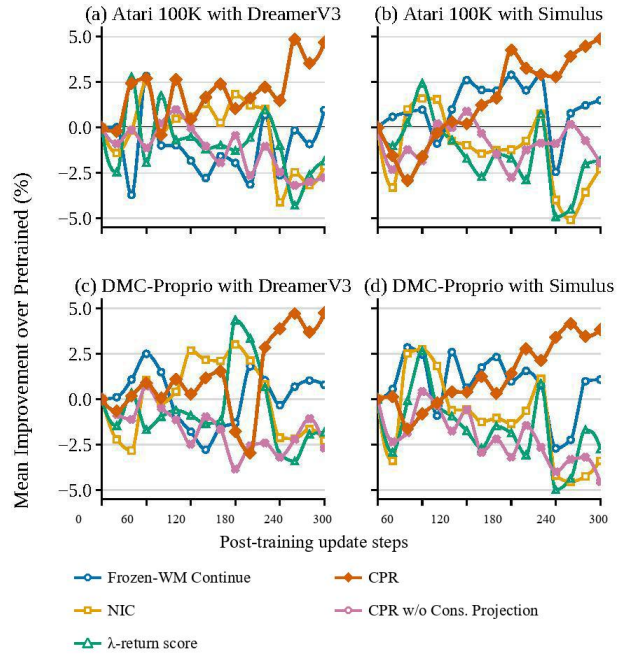


Figure 5: **Five-seed post-training improvement over Source for CPR and its component variants on Atari and DMC.** NIC keeps CPR’s branch budget but removes comparative weights; the λ -return variant replaces the Q -based branch score; and CPR w/o Conservative Projection removes the source-constrained update.

local ordering needed for repair. The horizon-dependent decay, especially on MT50, motivates using short branches and treating the model as a comparator rather than as an unconstrained long-horizon simulator.

Figure 5 isolates CPR’s three components. NIC and Frozen-WM Continue are weaker and less stable, so extra actor updates or imagined branches alone do not explain the gains. Replacing the Q -based score with λ -returns reduces performance in all four settings, while removing the conservative update causes late-stage instability as the policy moves away from source-policy support. In our experiments, stable repair therefore requires both action-specific branch ranking and source-constrained policy updates.

The ablations target different alternative explanations. Frozen-WM Continue tests whether the final world model merely needs more native actor-critic updates after source training. NIC uses the same replay anchors and branch budget as CPR but removes the branch score, so it tests whether exposing the actor to more imagined first actions is sufficient. The λ -return variant keeps CPR’s sampling and actor objective but ranks candidates with the standard bootstrapped branch return, testing whether ordinary finite-horizon targets can replace the action-conditioned score. Finally, removing the conservative update tests whether local ranking alone remains reliable once the actor is allowed to move away from the source-policy neighborhood. The combined pattern is consistent across Atari and DMC: useful repair requires a

Table 3: **Branch-score and Q -evaluator ablations under matched sampling, actor optimization, and compute.** Entries report mean relative gain over Source. Q -only keeps the evaluator and conservative actor update but removes the imagined branch correction.

Variant	Atari 100K		DMC-Proprio	
	DV3	Simulus	DV3	Simulus
λ -return score	-2.53	-2.01	-2.48	-2.79
Q -only	+1.82	+0.33	+1.07	-1.54
Full CPR	+4.70	+5.15	+4.78	+3.84

local ordering signal and an update rule that keeps selected actions on source-policy support.

The unstable no-projection curves are also informative. They do not show that the local ranking signal is useless; rather, they show that repeatedly fitting locally preferred actions can move the actor into regions where those rankings were not validated. CPR’s source-likelihood weighting and KL term act as a support filter for the comparative signal. They keep the update close to the behavior that generated the replay anchors, which is exactly the regime tested by the real-versus-model diagnostics.

Appendix *Additional Ablations* further compares fixed π_0 -continuation with moving π_θ -continuation in the candidate branches; the fixed continuation isolates the first-action counterfactual and avoids coupling branch scores to a drifting future policy. Task-level failures and possible causes, including unreliable local rankings and tasks requiring rare events or longer-horizon exploration, are analyzed in Appendix *Task-Level Patterns and Failure Modes* and Appendix *Limitations*.

Are the gains explained by branch returns or the new Q head? Table 3 rules out the new Q head as the main source of improvement. Q -only produces small and inconsistent changes, while λ -return scoring is negative in all four settings. Full CPR performs best under matched branch sampling, actor optimization, and compute. Thus the gain is not explained by simply adding an action-conditioned evaluator, nor by reusing standard branch returns with CPR’s regression loss; it comes from using the evaluator to anchor same-state branch comparisons before applying the actor update.

Related Work

Policy refinement with pretrained world models. Recent studies explore improving policies after world-model training without restarting the full RL pipeline. DiWA (Chandra et al. 2025) adapts diffusion policies using an offline world model; World4RL (Jiang et al. 2025) refines robotic manipulation policies inside a frozen diffusion world model; WPT (Jiang et al. 2026) distills world-model-guided teacher behavior into lightweight policies; and PWM (Georgiev et al. 2025) extracts continuous-control policies from pretrained multi-task world models through first-order optimization. These methods typically rely on complete imagined trajectories, teacher policies, or differentiable model objectives as the supervision signal. CPR instead extracts local policy-improvement

signals from short same-state branches, where candidate actions differ only at the first step and later actions follow the frozen source policy.

Planner-to-policy distillation. Another line of work improves policies by distilling stronger action-selection mechanisms into neural actors. Guided Policy Search (Levine and Koltun 2013), POPLIN (Wang and Ba 2019), MPDP (Li et al. 2023), MuZero reanalysis (Schrittwieser et al. 2020), ROSMO (Liu et al. 2023), TD-MPC2 (Hansen, Su, and Wang 2024), and Newt (Hansen, Su, and Wang 2025) use trajectory optimization, search, or planning-based targets to guide policy learning. These approaches generally require an explicit planner or optimize over multi-step trajectories. CPR avoids a separate planner teacher: it performs lightweight same-state comparisons between alternative first actions inside the frozen world model and directly updates the actor with the resulting relative advantages.

Preference-based policy updates. CPR is also related to preference-based optimization methods that learn from relative comparisons rather than absolute returns. DPO (Rafailov et al. 2023) learns from chosen–rejected pairs, while GRPO-style methods (Shao et al. 2024; Guo et al. 2025) compare multiple responses sampled from the same prompt. Recent RL approaches further derive preference signals from imagined futures: WMPO (Zhu et al. 2026) applies GRPO to imagined VLA trajectories, DreamPlan (Jia et al. 2026) fine-tunes planners from video-world-model rollouts, and CRAFT (Chen et al. 2026) builds advantages from candidate future outcomes with grounded residual correction. These methods typically compare complete responses, plans, or trajectories. CPR focuses on local action selection: it compares alternative first actions from the same latent state under a fixed source-policy continuation and performs KL-regularized weighted regression without backpropagating through the world model.

Conclusion

We presented Comparative Policy Repair (CPR), a lightweight post-training framework that improves completed world-model agents without additional environment interaction or retraining the world model. Our results demonstrate that frozen world models, although imperfect for long-horizon prediction, can preserve useful local action-ranking information when combined with an action-conditioned evaluator and a source-constrained policy update. Across Atari 100K, DMC-Proprio, and Meta-World MT50, CPR consistently improves strong pretrained agents with only a small fraction of the original training cost. Further analysis shows that the improvement arises from reliable local comparisons among alternative actions rather than from additional optimization alone. These findings suggest a complementary role for world models beyond online training: they can serve as reusable computational substrates for post-training diagnosis and policy refinement. Future work may explore more adaptive repair mechanisms and broader policy improvement strategies while addressing the challenges of distribution shift and long-horizon generalization.

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